Looping Statements

--> If you want to execute certain statements multiple times then use looping statements
--> If you know number of iterations then use for-loop
--> If you don't know number of iterations, but you need to execute multiple times based on condition then use while loop

for loop

--> for loop we can use in 2 ways
1. Based on group of elements
2. Based on range(-,-,-) function

for loop Based on group of elements

--> In a group of elements, if you want to bring elements one by one then use for loop

Syntax:
```
for variable in group:
    Statement1-1
    ........................
Statement-2
```

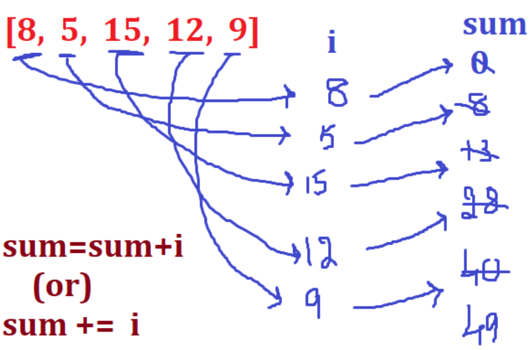
Ex: Consider a = [8, 5, 15, 12, 9], Now print elements one by one

```python
a = [8,  5,  15,  12,  9]
for i in a:
    print(i)
print('Thank you')
```

Ex: Consider a = [8, 5, 15, 12, 9], Now display even numbers

```python
a = [8,  5,  15,  12,  9]
for i in a:
    if i%2==0:
        print(i)
print('Thank you')
```

Ex: Consider a = [8, 5, 15, 12, 9], Find sum of all numbers



sum=sum+i
(or)
sum += i

```python
a = [8,  5,  15,  12,  9]
sum = 0
for i in a:
    sum += i
print(sum)
```

Ex: Consider a = [8, 5, 15, 12, 9], Find sum of even, odd numbers

Even number sum is : 20
Odd numbers sum is : 29

```
a = [8,  5,  15,  12,  9]
esum,osum = 0,0
for i in a:
    if i%2==0:
        esum += i
    else:
        osum += i
print('Even Sum is : ',esum)
print('Odd sum is  : ',osum)
```

range( ) function

--> range( ) function is used to generate sequence of numbers based on input
--> range( ) function take 3 input

1. start value --> Default value is 0
2. stop value
3. step value --> Default value is 1

Syntax: range(start, stop, step)

Ex: range(1, 6, 1) --> 1 2 3 4 5
range(1, 6, 2) --> 1 3 5
range(5, 1, -1) --> 5 4 3 2
range(1, 6) --> range(1, 6, 1) ---> 1 2 3 4 5
range(5) --> range(0, 5, 1) ---> 0 1 2 3 4

--> W.a.p to print 1 to 5 numbers

1 2 3 4 5

```
for i in range(1,6):
    print(i)
```

i
~~1~~
~~2~~
~~3~~
~~4~~
5

print(i, end='\n')

1
2
3
4
5

```
for i in range(1,6):
    print(i,end='\t')
```

--> W.a.p to read n value then print numbers from 1 to n

```
n = int(input('Enter any Number : '))
for i in range(1,n+1):
    print(i)
```

--> W.a.p to read n value then find sum of all even numbers b/w 1 to n

Enter n value : 5
Sum of Even numbers : 6 (2+4)

```
n = int(input('Enter any Number : '))
sum = 0
for i in range(1,n+1):
    if(i%2==0):
        sum += i
print(sum)
```

Factorial

-->Product of 1 to n or n to 1 is called as Factorial of a given number

Ex: n =5
factorial is : 120 ---> (1 * 2 * 3 * 4 * 5)

-->W.a.p to read a number then find factorial of a given number

```
n = int(input('Enter any Number : '))
fact = 1
for i in range(1,n+1):
    fact = fact * i
print('Factorial is : ',fact)
```

-->W.a.p to read n value then print all divisibles for that number

Enter any Number : 6

6 % 1 == 0 ✓
6 % 2 == 0 ✓
6 % 3 == 0 ✓
6 % 4 == 0 ✗
6 % 5 == 0 ✗
6 % 6 == 0 ✓

n % i == 0

Enter any Number : 5

5 % 1 == 0 ✓
5 % 2 == 0 ✗
5 % 3 == 0 ✗
5 % 4 == 0 ✗
5 % 5 == 0 ✓

n % i == 0

```
n = int(input('Enter any Number : '))
for i in range(1,n+1):
    if n%i==0:
        print(i)
```

1) 5 (5
5

0

-->W.a.p to read n value then count no.of divisibles for that number

Enter any Number : 6

6 % 1 == 0 ✓
6 % 2 == 0 ✓
6 % 3 == 0 ✓
6 % 4 == 0 ✗
6 % 5 == 0 ✗
6 % 6 == 0 ✓

count
0
1
2
3
4

n % i == 0

Total No.of Divisibles are: 4

```
n = int(input('Enter any Number : '))
count = 0
for i in range(1,n+1):
    if n%i==0:
        count += 1
print('Total No.of Divisibles are : ',count)
```

prime Number :

--> Any number which has only two divisibles i.e. 1 and itself then it is called as prime number

Ex: 2, 3, 5, 7, 11, 13, 17, ...

--> W.a.p to read a number then check given number is Prime or not

--> Read n value
--> find no.of divisibles
--> check count==2 or not
    if yes then print 'Prime number'
    otheriwse print 'Not a prime number'

Ex:

```
n = int(input('Enter any Number : '))
count = 0
for i in range(1,n+1):
    if n%i==0:
        count += 1
if count==2:
    print('Given Number is Prime Number')
else:
    print('Given Number is not a Prime Number')
```